

Net2_Site-to-site VPN - SKAL afleveres

Step 1.2:

```
R1(config)# license boot module c1900 technology-package securityk9
```

Aktiverer SecurityK9-sikkerhedspakken (kryptografi og avancerede sikkerhedsfunktioner) på en Cisco 1900-serie router ved næste genstart.

Ved at køre show version kan man se det på routeren:

```
-----  
Device#      PID                SN  
-----  
*0           CISCO1941/K9           FTX1524F8G8  
  
Technology Package License Information for Module:'c1900'  
  
-----  
Technology    Technology-package    Technology-package  
Current       Type                 Next reboot  
-----  
ipbase        ipbasek9              Permanent          ipbasek9  
security      securityk9            Evaluation         securityk9  
data          disable               None               None  
  
Configuration register is 0x2102
```

Step 1.3:

```
R1(config)# access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

Laver en extended access-list der tillader alt fra netværket 192.168.1.0 til 192.168.3.0 netværket.

```
R1#show access-list 110  
Extended IP access list 110  
    permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

Step 1.4:

```
R1(config)# crypto isakmp policy 10
```

Definerer en IKE Phase 1 sikkerhedspolitik med prioritet 10.

```
R1(config-isakmp)# encryption aes 256
```

Sætter aes encryption op med 256 bit encryption

```
R1(config-isakmp)# authentication pre-share
```

Sætter måden enhederne autentikere med hinanden på, specifikt med en pre-shared key.

```
R1(config-isakmp)# group 5
```

Sætter et dh-group nummer som kontrolere stykren af nøgle udvekslingen.

```
R1(config-isakmp)# exit
```

```
R1(config)# crypto isakmp key vpnpa55 address 10.2.2.2
```

Opsætter isakmp op med nøglen vpnpa55 og angiver IP-adressen for den eksterne VPN-partner, som denne specifikke nøgle gælder for.

Step 1.5:

```
R1(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
```

Den laver et navngivet sæt, som hedder (VPN-Sæt) af de algoritmer der beskytter dataen, den bruger esp-aes med AES til at encryptere dataen, og så bruger den SHA-HMAC til integritet af pakken

```
R1(config)# crypto map VPN-MAP 10 ipsec-isakmp
```

Laver et crypto map navngivet VPN-MAP, med sekvens nummeret 10.

```
R1(config-crypto-map)# description VPN connection to R3
```

Laver en description.

```
R1(config-crypto-map)# set peer 10.2.2.2
```

Sætter den fjerne enhed entrien applikeres til.

```
R1(config-crypto-map)# set transform-set VPN-SET
```

Definer hvilket transform-set der skal bruges

```
R1(config-crypto-map)# match address 110
```

Siger den skal matche adressen 110, som er ACL'en i step 3.

```
R1(config-crypto-map)# exit
```

Step 1.6:

```
R1(config)# interface s0/0/0
```

```
R1(config-if)# crypto map VPN-MAP
```

Binder crypto mappet VPN-MAP til interfacet s0/0/0.

Step 2.1:

Show version på R3.

```
-----  
Device#      PID                SN  
-----  
*0           CISC01941/K9          FTX1524I27D  
  
Technology Package License Information for Module:'c1900'  
  
-----  
Technology    Technology-package    Technology-package  
              Current      Type              Next reboot  
-----  
ipbase        ipbasek9             Permanent         ipbasek9  
security      securityk9           Evaluation        securityk9  
data          disable              None              None  
  
Configuration register is 0x2102
```

Step 2.2: Samme kommando som step 1.3.

```
R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```

Step 2.3: Samme kommandoer som step 1.4.

```
R3(config)# crypto isakmp policy 10  
R3(config-isakmp)# encryption aes 256  
R3(config-isakmp)# authentication pre-share  
R3(config-isakmp)# group 5  
R3(config-isakmp)# exit  
R3(config)# crypto isakmp key vpnpa55 address 10.1.1.2
```

Step 2.4: Samme kommandoer som step 1.5.

```
R3(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac  
R3(config)# crypto map VPN-MAP 10 ipsec-isakmp  
R3(config-crypto-map)# description VPN connection to R1  
R3(config-crypto-map)# set peer 10.1.1.2  
R3(config-crypto-map)# set transform-set VPN-SET  
R3(config-crypto-map)# match address 110  
R3(config-crypto-map)# exit
```

Step 2.5: Samme kommandoer som step 1.6.

```
R3(config)# interface s0/0/1  
R3(config-if)# crypto map VPN-MAP
```

Step 3.1:

Show crypto ipsec sa

Viser ipsec security associations information.

```
R1#show crypto ipsec sa
```

```
interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 10.1.1.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
    current outbound spi: 0x0(0)

  inbound esp sas:

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:

  outbound ah sas:

  outbound pcp sas:
```

Step 3.2:

Ping – kommando brugt til at troubleshoot forbindelse med ICMP.

Step 3.3:

Show crypto ipsec sa EFTER PC-C ping fra PC-A.

```
R1#show crypto ipsec sa
```

```
interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 10.1.1.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.2 port 500
    PERMIT, flags={origin is acl.}
    #pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
    #pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

  local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
  current outbound spi: 0xF8CC82AF(4174152367)
```

```
inbound esp sas:
  spi: 0x84177223(2216129059)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2007, flow_id: FPGA:1, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec): (4525504/3578)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE
```

```
inbound ah sas:
```

```
inbound pcp sas:
```

```
outbound esp sas:
  spi: 0xF8CC82AF(4174152367)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2008, flow_id: FPGA:1, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec): (4525504/3578)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE
```

```
outbound ah sas:
```

```
outbound pcp sas:
```

Step 3.4:

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=6ms TTL=126
Reply from 192.168.2.3: bytes=32 time=48ms TTL=126
Reply from 192.168.2.3: bytes=32 time=4ms TTL=126
Reply from 192.168.2.3: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 48ms, Average = 16ms
```

Step 3.5:

Nummeret af pakker ændres ikke efter ping fra PC-A til PC-B, det beviser trafikken ikke bliver enkrypteret.

```
R1#show crypto ipsec sa

interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 10.1.1.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.2 port 500
    PERMIT, flags={origin is acl,}
    #pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
    #pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0
```